

Mitchell C. Elliott

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Santa Cruz, CA 95060

SKILLS

Languages and Frameworks: Python, C, C++, Java, JavaScript, Bash, SQL, React, Node.js, Flask, FastAPI, PyTorch, Pandas
Tools and Platforms: Docker, Kubernetes, AWS (EC2, S3, RDS, Lambda), Linux, Git, Jenkins, Jira, New Relic, Kafka, Terraform
Databases and Storage Systems: PostgreSQL, pgvector, MySQL, Redis, OpenSearch, MinIO, Ceph, SQLite, MongoDB, Cassandra

EXPERIENCE

University of California, Santa Cruz Jan. 2026 – Present
Research and Development Engineer Santa Cruz, CA

- Leading a 3-person research team developing sustainable storage-system modeling tools for data-center design
- Designing sustainable hybrid storage architectures with Prof. Abel Souza to reduce end-to-end carbon impact in data centers
- Developing simulation tools to model carbon emissions, performance impacts, and tradeoffs of storage system configurations

Research Affiliate Jan. 2024 – Jan. 2026

- Researched sustainable data-center storage systems with a focus on carbon, cost, and performance tradeoffs
- Applied life cycle assessment (LCA) methods to compare storage media by energy use and environmental footprint
- Developed simulation tools for evaluating storage-system configurations under different workload and design assumptions

Teaching Assistant Jan. 2023 – Dec. 2024

- Served as the Head TA for CSE 130: Principles of Computer System Design across 6 quarters; mentored 1000+ students
- Conducted group discussion sections, created assignments and quizzes, proctored exams, and managed course infrastructure
- Created and maintained an automated CI/CD pipeline for grading programming assignments on GitLab

Perfectly Snug July 2023 – Sept. 2023
Software Engineer Intern Remote

- Developed a Material Requirements Planning (MRP) system with BOM management using Python, MySQL, Node.js, and React
- Automated part tracking and manufacturing workflow management to reduce manual effort and production delays
- Built dynamic dashboards and analytics pipelines using Shopify API data to optimize daily build plans and inventory decisions

ParkourSC, Inc. July 2022 – Sept. 2022
DevOps Engineer Intern Remote

- Redesigned and deployed alert condition policies across multiple Kubernetes clusters in New Relic to improve signal quality
- Eliminated false positives, reducing noise in incident queues and increasing engineering focus on high-priority alerts
- Automated cluster alert muting during upgrades by integrating New Relic's NerdGraph API with Jira workflows

PROJECTS

Large-Scale Data Infrastructure Platform Jan. 2026 – Present

- Built reproducible data pipelines to ingest, organize, and process large-scale Google cluster and power datasets
- Developed DuckDB and Parquet workflows to enable scalable querying, schema mirroring, and repeatable analytics
- Created reusable CLI, testing, and ingestion tooling for bounded experiments and simulator-ready workload analysis

Data Center Carbon TCO Simulator Jan. 2024 – Present

- Created a Python-based simulator that models end-to-end total cost of ownership (TCO) for data centers
- Developed a multi-metric optimization framework integrating cost, performance, and sustainability to guide tiered storage layouts
- Identified energy-efficient hybrid storage designs using workload-aware simulations and access pattern profiling

EDUCATION

University of California, Santa Cruz *M.S. Computer Science* | Sept. 2022 – Dec. 2024
B.S. Computer Science | Sept. 2018 – June 2022